

ZF Multipliers

Thijs Lenssen

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1 Slope restricted uncertainty (Soft IQC) [3]

This document concerns the hard factorization of Zames-Falb(ZF) type multipliers. More specifically, we will consider the multipliers that describe the class of incremental sector conditions. Consider the nonlinear operator $w(t) = \Delta(z(t))$ for $t \geq 0$, $z \in L_2$, where $\Delta : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\Delta(0) = 0$ as well as the incremental sector condition

$$0 \leq \frac{\Delta(x_1) - \Delta(x_2)}{x_1 - x_2} \leq \sup_{x_1 \neq x_2} \frac{\Delta(x_1) - \Delta(x_2)}{x_1 - x_2} < \alpha \quad (1)$$

for all $x_1 \neq x_2$. The fixed constant $0 < \beta$ can be viewed as the bounds on the slope of the nonlinearities; due to $0 \in [\alpha, \beta]$ it is assured that $\tau\Delta$ shares these properties with Δ for all $\tau \in [0, 1]$. Since $\Delta(0) = 0$ we note that Δ also satisfies the sector condition

$$(\Delta(x) - \alpha x)(\beta x - \Delta(x)) \geq 0, \quad \forall x \in \mathbb{R} \quad (2)$$

Class 1. Suppose that $\Delta : \mathbb{R} \rightarrow \mathbb{R}$ with $\Delta(0) = 0$ satisfies the incremental sector condition (1) for $0 < \alpha$, and let Δ denote the induced memoryless operator. Suppose that $h \in L_1(0, \infty)$ satisfies $h(t) \geq 0$ a.e. and $\|h\|_1 < 1$, and define

$$(Hx)(t) := \int_0^\infty h(\tau)x(t - \tau) d\tau, \quad M := I - H, \quad (3)$$

where $x(t) = 0$ for $t < 0$. Then, for every $z \in L_2$, the corresponding Zames-Falb multiplier admits the soft factorization below,

$$\left\langle \frac{1}{\sqrt{2}} \begin{bmatrix} M & 1 \\ M & -1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{\alpha} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ \Delta(z) \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} M & 1 \\ M & -1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{\alpha} \\ 0 & I \end{bmatrix} \begin{bmatrix} z \\ \Delta(z) \end{bmatrix} \right\rangle \geq 0. \quad (4)$$

Extending in a similar way to $w(t, s) = \Delta(z(t, s))$ for $t \geq 0$ and $s \in [a, b]$, such that (1) also holds for $w(t, s), z(t, s)$ we can find the following class of zames-falb multipliers

Class 2. Suppose that $\Delta : \mathbb{R} \rightarrow \mathbb{R}$ with $\Delta(0) = 0$ satisfies the incremental sector condition (1) for $0 < \beta$, and let Δ denote the induced memoryless operator. Suppose that $h \in L_1[0, \infty) \times [a, b]$, satisfies $h(t, s) \geq 0$ a.e. and $\|h\|_1 < 1$, and define

$$(Hx)(t, s) := \int_0^\infty h(\tau, s)x(t - \tau, s) d\tau, \quad M := I - H, \quad (5)$$

where $x(t, s) = 0$ for $t < 0$. Then, for every $z \in L_2[0, \infty) \times [a, b]$, the corresponding Zames-Falb multiplier admits the soft factorization below,

$$\left\langle \frac{1}{\sqrt{2}} \begin{bmatrix} M & 1 \\ M & -1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{\alpha} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ \Delta(z) \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} M & 1 \\ M & -1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{\alpha} \\ 0 & I \end{bmatrix} \begin{bmatrix} z \\ \Delta(z) \end{bmatrix} \right\rangle \geq 0. \quad (6)$$

Note that $h(t, s)$ acts pointwise in the spatial variable s , and therefore introduces no coupling between distinct spatial locations. In a PIE realization, this corresponds to state-space operators that are spatially local, i.e., multiplication operators in s . Consequently, the integral-kernel components of the associated 3-PI operators vanish, so that $R_1 = R_2 = 0$ with only the R_0 component remaining. A simple construction of such $h(t, s)$ is to take $h(t, s) = h_{ODE}(t) \otimes I_{\mathcal{L}(L_2[a, b])}$, where $h_{ODE}(t)$ is a scalar function that satisfies the conditions of the first class. This corresponds to the following 3-PI system

$$H_{\mathcal{L}(L_2[0, \infty) \times [a, b])} := \begin{cases} \mathcal{T} = \mathcal{A}\mathbf{x}(t) + \mathcal{B}w(t), \\ z(t) = \mathcal{C}\mathbf{x}(t), \end{cases}$$

Then we define the 3-PI operators of H as $\mathcal{T} = I$, $(\mathcal{A}x)(s, t) = Ax(s, t)$, $(\mathcal{B}w)(s, t) = Bw(s, t)$, and $(\mathcal{C}x)(s, t) = Cx(s, t)$, where A, B, C are the corresponding matrices of $h_{ODE}(t)$.

1.1 Spatial Lifting of IQC

We denote the extended L_2 space as $L_{2,e}$, let $\mathcal{F}$ denote the set of hard factorizations. We say the uncertainty $\Delta : L_{2,e} \rightarrow L_{2,e}$ satisfies the hard iqc

$$\left\langle p_T \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix}, p_T V \Psi \begin{bmatrix} z \\ \Delta z \end{bmatrix} \right\rangle_{L_2} \geq 0 \quad (7)$$

for all $T \geq 0$, $z \in L_{2,e}$, and hard factorizations $(\Psi, V) \in \mathcal{F}$, with $\Psi : L_{2,e}^n \rightarrow L_{2,e}^{\tilde{n}}$ and $V \in \mathbb{R}^{\tilde{n} \times \tilde{n}}$. Now we extend Δ pointwise over the spatial domain $[a, b]$. Denote the signal space $L_{\mathbf{H},e} = L_2([0, T] \times [a, b])$ for any signal $u \in L_{\mathbf{H},e}$

$$\langle u, u \rangle_{L_{\mathbf{H},e}} = \int_0^T \int_a^b |u(t, s)|^2 ds dt < \infty, \quad \forall T \geq 0 \quad (8)$$

Define the lifted uncertainty operation, $\hat{\Delta} : L_{\mathbf{H},e} \rightarrow L_{\mathbf{H},e}$, with $(\hat{\Delta}v)(s) = \Delta v(s)$, i.e. Δ applied at every $s \in [a, b]$, the lifted hard factorization $(\hat{\Psi}, \hat{V})$, with $\hat{\Psi} : L_{\mathbf{H},e} \rightarrow L_{\mathbf{H},e}$ and $\hat{V} : L_2[a, b] \rightarrow L_2[a, b]$, such that $(\hat{\Psi}(s), \hat{V}(s)) \in \mathcal{F}$ for all $s \in [a, b]$.

Corollary 1. *If (7) holds for all $T > 0$ and $z \in L_{2,e}[0, \infty)$, then*

$$\left\langle p_T \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix}, p_T \hat{V} \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix} \right\rangle_{L_{\mathbf{H}}} \geq 0 \quad (9)$$

for all $u \in L_{\mathbf{H},e}$ and $T \geq 0$

Proof. Assume (7) is satisfied. We need to show that (9) is also satisfied. First examine that

$$\left\langle p_T \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix}, p_T \hat{V}(s) \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix} \right\rangle_{L_2} \geq 0, \quad \forall s \in [a, b]$$

Thus by integrating over s we find

$$\int_a^b \left\langle p_T \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix}, p_T \hat{V}(s) \hat{\Psi}(s) \begin{bmatrix} u(s) \\ \Delta u(s) \end{bmatrix} \right\rangle_{L_2} ds = \left\langle p_T \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix}, p_T \hat{V} \hat{\Psi} \begin{bmatrix} u \\ \hat{\Delta}u \end{bmatrix} \right\rangle_{L_{\mathbf{H}}} \geq 0$$

□

2 Positive Negative multipliers yield hard factorizations [2]

The main result of [2] is the establishment of

1. (Ψ, M) hard factorizations.
2. $P = P^\top$ is a solution to the KYP lemma. (no enforcement of $P > 0$)

for the class of postive-negative multipliers. Before stating the theorem we need to define J -spectral factors.

Definition 1. (Ψ, M) is called a $J_{n,m}$ -spectral factor of $\Pi = \Pi^\sim \in \mathbb{RL}_\infty^{(n+m) \times (n+m)}$ if $\Pi = \Psi^\sim M \Psi$, $M - \begin{bmatrix} I_n & 0 \\ 0 & -I_m \end{bmatrix}$, and $\Psi, \Psi^{-1} \in \mathbb{RH}_\infty^{(n+m) \times (n+m)}$

Using J -spectral factorizations, Seiler [2] derives the following theorem.

Theorem 2. Let $\Pi = \Pi^\sim$ and partition as $\begin{bmatrix} \Pi_{11} & \Pi_{21} \\ \Pi_{21}^\sim & \Pi_{22} \end{bmatrix}$ where $\Pi_{11} \in \mathbb{RL}_\infty^{n \times n}$ and $\Pi_{22}^{m \times m}$. If $\Pi_{11}(j\omega) > 0$ and $\Pi_{22}(j\omega) < 0$, i.e. Π is strictly positive-negative(PN), $\forall \mathbb{R} \cup \{\infty\}$, then

1. Π has a $J_{n,m}$ -spectral factorization (Ψ, M) .
2. The $J_{n,m}$ -spectral factorization (Ψ, M) is a hard factorization of Π .
3. For $G \in \mathbb{RH}_\infty^{n \times m}$, let $(\hat{A}, \hat{B}, \hat{C}, \hat{D})$ denote the state-space realization of $\Psi \begin{bmatrix} G \\ I \end{bmatrix}$. All solutions $P = P^\top$ to the KYP LMI satisfy $P \geq 0$.

References

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